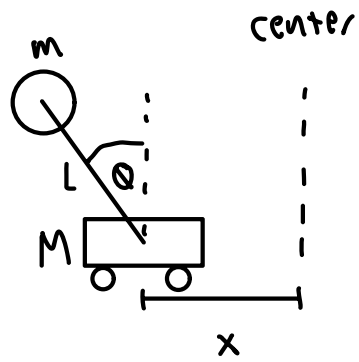

$$x_c = x \rightarrow \dot{x}_c = \dot{x}$$
$$T_c = \frac{1}{2} M \dot{x}^2$$

$$x_p = x + L\sin(\theta)$$
$$\dot{x}_p = \dot{x} + L\dot{\theta}\cos(\theta)$$
$$\dot{x}_p^2 = \dot{x}^2 + 2L\dot{\theta}\cos(\theta) + (L\dot{\theta}\cos\theta)^2$$
$$y_p = L\cos\theta \rightarrow \dot{y}_p = -L\dot{\theta}\sin(\theta)$$
$$\dot{y}_p^2 = L^2\dot{\theta}^2\sin^2\theta$$
$$T_p = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m(\dot{x}^2 + 2L\dot{x}\dot{\theta}\cos\theta + L^2\dot{\theta}^2)$$

$$U = mg\,y_p = mgL\cos\theta$$

$$\mathcal{L} = T - U = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m(\dot{x}^2 + 2L\dot{x}\dot{\theta}\cos\theta + L^2\dot{\theta}^2 - 2gL\cos\theta)$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = Q$$

$$\theta: \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{L}}{\partial \theta} = 0 \Rightarrow \frac{m}{2}\left(\frac{d}{dt}\left(2L\dot{x}\cos\theta + 2L^2\dot{\theta}\right) - 2gL\sin\theta + 2L\dot{x}\dot{\theta}\sin\theta\right) = 0$$

$$\Rightarrow \frac{m}{2}\left(2L(\ddot{x}\cos\theta - \dot{x}\dot{\theta}\sin\theta) + 2L^2\ddot{\theta} - 2gL\sin\theta + 2L\dot{x}\dot{\theta}\sin\theta\right) = 0$$

$$\Rightarrow mL\ddot{x}\cos\theta + mL^2\ddot{\theta} - mgL\sin\theta = 0$$

$$x: \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) - \frac{\partial \mathcal{L}}{\partial x} = Q \Rightarrow \frac{d}{dt}\left(M\dot{x} + \frac{1}{2}m\left(2\dot{x} + 2L\dot{\theta}\cos\theta\right)\right) = Q$$

$$\Rightarrow M\ddot{x} + m\ddot{x} + mL(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta) = Q$$

$$\Rightarrow \ddot{x}(M+m) + mL(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta) = Q$$

$$mL\ddot{x}\cos\theta + mL^2\ddot{\theta} - mgL\sin\theta = 0$$

$$\ddot{x}(M+m) + mL(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta) = Q$$

$$\Longrightarrow$$

$$\ddot{x}\cos\theta + L\ddot{\theta} = g\sin\theta$$

$$(m+M)\ddot{x} + mL\ddot{\theta}\cos\theta = Q + mL\dot{\theta}^2\sin\theta$$

$$\Longrightarrow$$

$$\begin{bmatrix} \cos\theta & L \\ m+M & mL\cos\theta \end{bmatrix}\begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} g\sin\theta \\ Q + mL\dot{\theta}^2\sin\theta \end{bmatrix}$$

<u>Linearizing:</u>

$$A = \begin{bmatrix} \cos\theta & L \\ m+M & mL\cos\theta \end{bmatrix}, \quad B = \begin{bmatrix} g\sin\theta \\ Q + mL\dot{\theta}^2\sin\theta \end{bmatrix}, \quad \ddot{q} = \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix}$$

$$\ddot{q} = A^{-1}B$$

$$A^{-1} = \frac{1}{\det(A)}\begin{bmatrix} mL\cos\theta & -L \\ -m-M & \cos\theta \end{bmatrix} \qquad \det(A) = mL\cos^2\theta - L(m+M)$$

$$= \frac{1}{mL\cos^2\theta - L(m+M)}\begin{bmatrix} mL\cos\theta & -L \\ -m-M & \cos\theta \end{bmatrix}$$

$$\ddot{q} = \frac{1}{mL\cos^2\theta - L(m+M)}\begin{bmatrix} mL\cos\theta & -L \\ -m-M & \cos\theta \end{bmatrix}\begin{bmatrix} g\sin\theta \\ Q + mL\dot{\theta}^2\sin\theta \end{bmatrix}$$

$$= \frac{1}{mL\cos^2\theta - L(m+M)}\begin{bmatrix} mgL\cos\theta\sin\theta - QL - mL^2\dot{\theta}^2\sin\theta \\ -g(m+M)\sin\theta + Q\cos\theta + mL\dot{\theta}^2\sin\theta\cos\theta \end{bmatrix}$$

$$\ddot{q} = \begin{bmatrix} \dfrac{mgL\cos\theta\sin\theta - QL - mL^2\dot{\theta}^2\sin\theta}{mL\cos^2\theta - L(m+M)} \\ \dfrac{-g(m+M)\sin\theta + Q\cos\theta + mL\dot{\theta}^2\sin\theta\cos\theta}{mL\cos^2\theta - L(m+M)} \end{bmatrix}$$

$$y = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$$

$$\dot{y} = \begin{bmatrix} \dot{x} \\ \dfrac{mgL\cos\theta\sin\theta - QL - mL^2\dot{\theta}^2\sin\theta}{mL\cos^2\theta - L(m+M)} \\ \dot{\theta} \\ \dfrac{-g(m+M)\sin\theta + Q\cos\theta + mL\dot{\theta}^2\sin\theta\cos\theta}{mL\cos^2\theta - L(m+M)} \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{bmatrix}$$

$$\dot{y} \approx J|_{eq}\, y$$

$$J|_{eq} = \begin{bmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial \dot{x}} & \frac{\partial f_1}{\partial \theta} & \frac{\partial f_1}{\partial \dot{\theta}} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial \dot{x}} & \frac{\partial f_2}{\partial \theta} & \frac{\partial f_2}{\partial \dot{\theta}} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial \dot{x}} & \frac{\partial f_3}{\partial \theta} & \frac{\partial f_3}{\partial \dot{\theta}} \\ \frac{\partial f_4}{\partial x} & \frac{\partial f_4}{\partial \dot{x}} & \frac{\partial f_4}{\partial \theta} & \frac{\partial f_4}{\partial \dot{\theta}} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & b & 0 \end{bmatrix}$$

$$a = -\frac{mg}{M}$$

$$b = \frac{g(m+M)}{LM}$$

<u>With driving force:</u>

$$\dot{y} = Ay + BQ$$

$$B = \frac{\partial f_i}{\partial Q} = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ -\frac{1}{ML} \end{bmatrix}$$

$$Q = -Ky$$

$$\dot{y} = Ay - BKy = (A - BK)y$$

$$J = \int_0^\infty (y^T Z y + Q^T R Q)\, dt$$

Minimize this matrix to find poles, then design K s.t. (A-BK) has these poles.